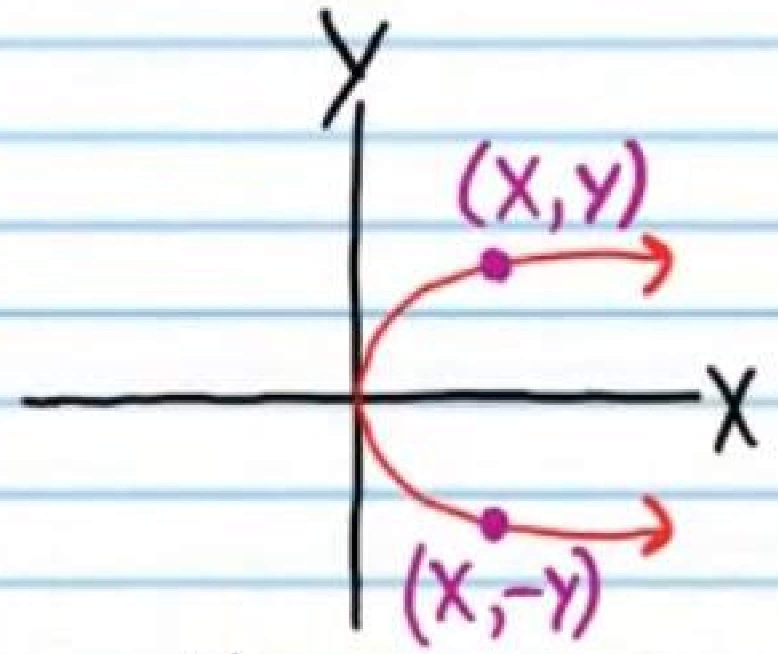
Symmetric Functions and Review on Exponents and Logarithms

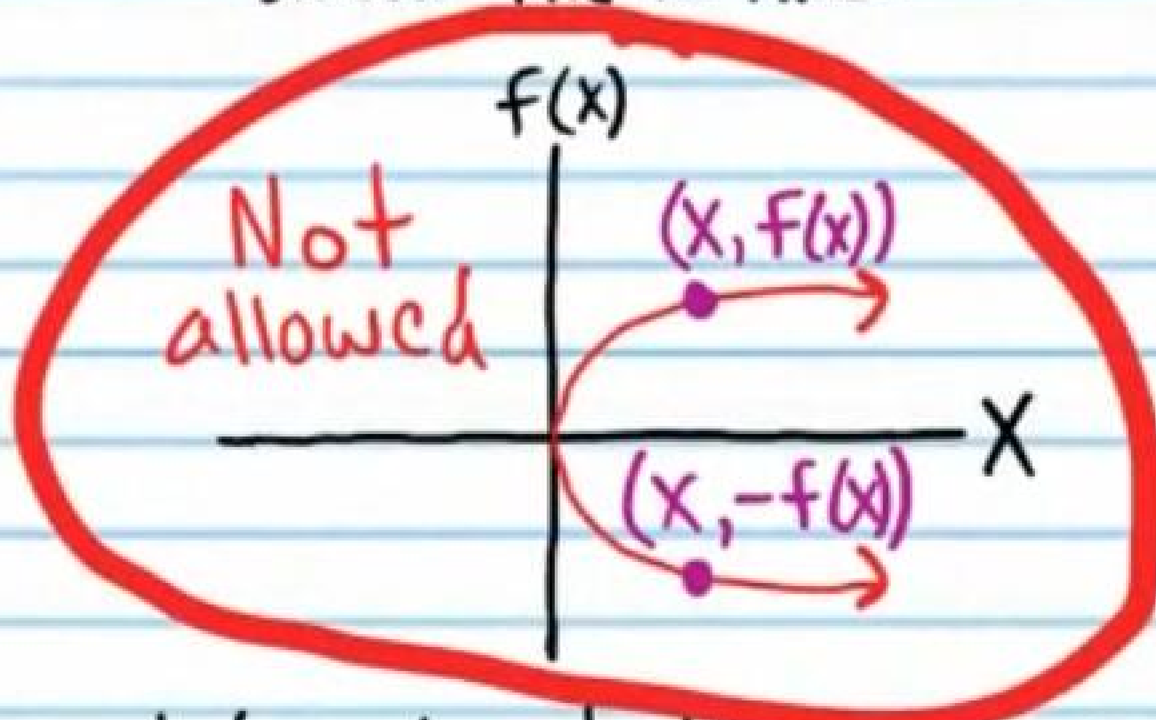
Symmetric Functions

Symmetry about the X-Axis

A Relation That Is Symmetric about the X-Axis



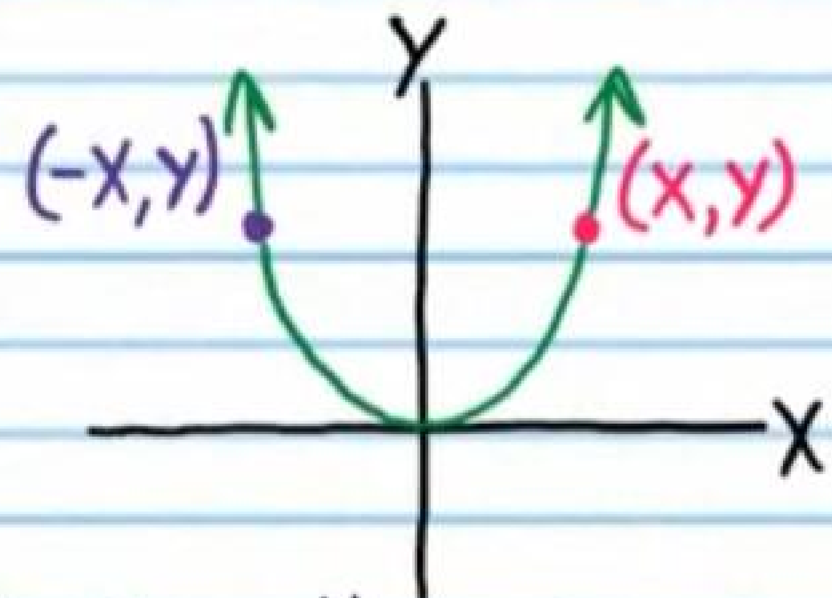
A ~~Function~~ That Is Symmetric about the X-Axis



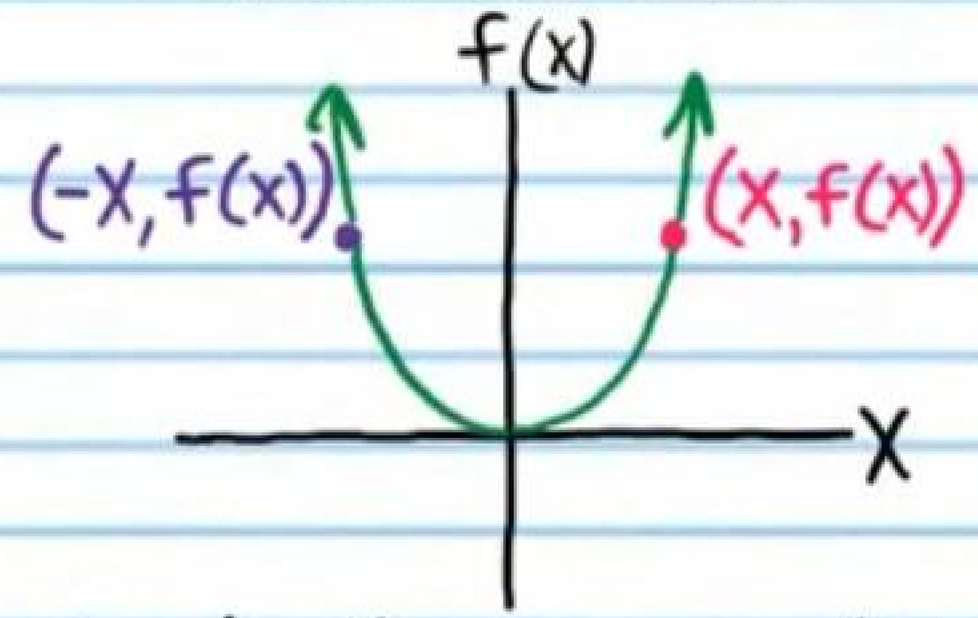
A function cannot be symmetric about the x-axis because for every input "x" there can be only one output f(x) according to the rules of functions.

Symmetry about the Y-Axis

A Relation That Is Symmetric about the Y-Axis



A Function That Is Symmetric about the Y-Axis



A function is symmetric about the y-axis if for every point $(x, f(x))$ on the graph of the function, there exists a point $(-x, f(x))$ that is also on the graph of the function.

Even Function: A function that is symmetric about the Y-axis.

Each table below has points collected from a graph of an even function. Fill the empty spaces with other points from the graph.

①

X	6	-8	5	7	-16	-6	8	-5	-7	16
f(x)	6	-1	2	10	-9	6	-1	2	10	-9

②

X	-1	5	-4	2	4	1	-5	4	-2	-4
f(x)	9	0	6	-2	8	9	0	6	-2	8

③ If "f" is an even function, and $f(14) = 7$, find $f(-14)$.

$(14, 7) \rightarrow (-14, 7)$ $\qquad f(-14) = \boxed{7}$

④ If "f" is an even function, and $f(-0.8) = 4.9$, find $f(0.8)$.

$(-0.8, 4.9) \rightarrow (0.8, 4.9)$ $\qquad f(0.8) = \boxed{4.9}$

⑤ If "f" is an even function, and $f(-3) = -19$, find $f(3)$.

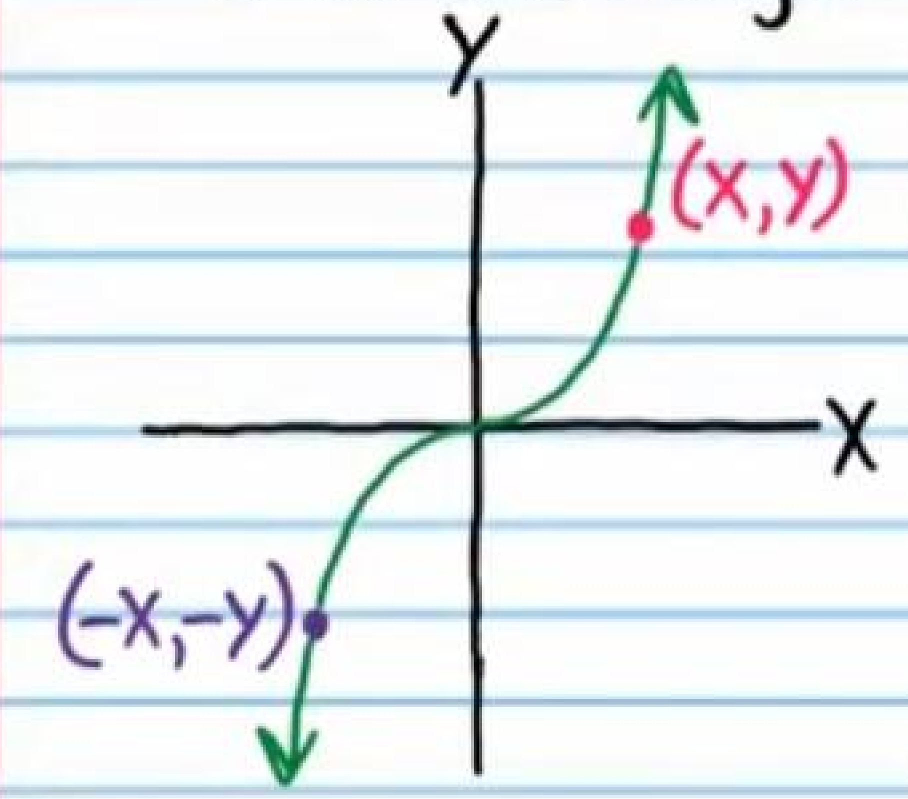
$(-3, -19) \rightarrow (3, -19)$ $\qquad f(3) = \boxed{-19}$

⑥ If "f" is an even function, and $f(\frac{6}{7}) = -1$, find $f(-\frac{6}{7})$.

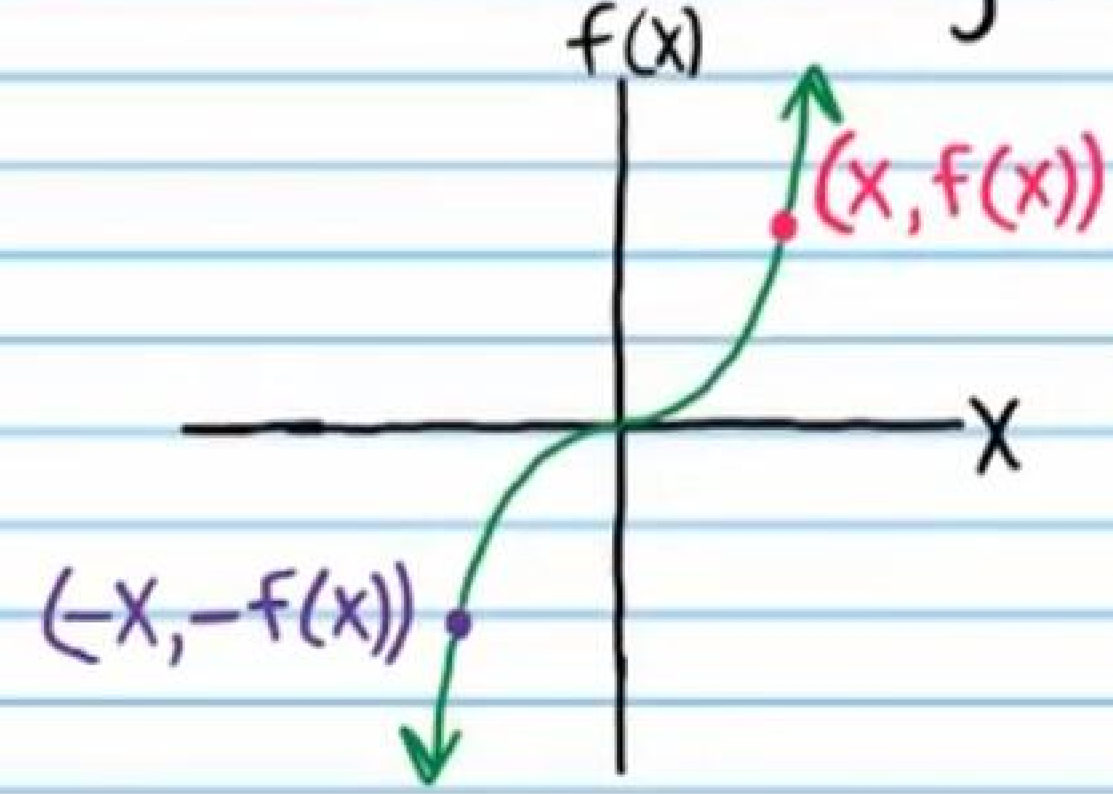
$f(-\frac{6}{7}) = \boxed{-1}$ $\qquad (\frac{6}{7}, -1) \rightarrow (-\frac{6}{7}, -1)$

Symmetry about the Origin

A Relation That Is Symmetric about the Origin



A Function That Is Symmetric about the Origin



A function is symmetric about the origin if for every point $(x, f(x))$ on the graph of the function, there exists a point $(-x, -f(x))$ that is also on the graph of the function.

Odd Function: A function that is symmetric about the origin.

Each table below has points collected from a graph of an odd function. Fill the empty spaces with other points from the graph.

⑦

X	-3	4	5	0	-10	3	-4	-5	0	10
f(x)	-5	-6	-1	0	10	5	6	1	0	-10

⑧

X	3	4	0	-77	10	-3	-4	0	77	-10
f(x)	88	9	-1	4	9	-88	-9	1	-4	-9

⑨ If "f" is an odd function, and $f(-9) = -20$, find $f(9)$.

$(-9, -20) \rightarrow (9, 20)$ $\quad f(9) = \boxed{20}$

⑩ If "f" is an odd function, and $f(-\frac{10}{9}) = \frac{1}{2}$, find $f(\frac{10}{9})$.

$(-\frac{10}{9}, \frac{1}{2}) \rightarrow (\frac{10}{9}, -\frac{1}{2})$ $\quad f(\frac{10}{9}) = \boxed{-\frac{1}{2}}$

⑪ If "f" is an odd function, and $f(2.4) = 5.9$, find $f(-2.4)$.

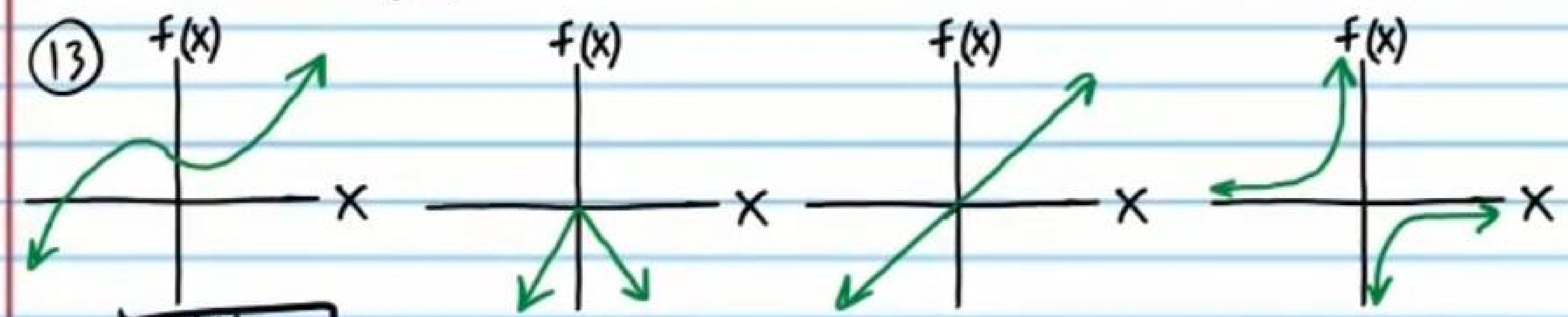
$(2.4, 5.9) \rightarrow (-2.4, -5.9)$ $\quad f(-2.4) = \boxed{-5.9}$

⑫ If "f" is an odd function, and $f(-8) = -6$, find $f(8)$.

$(-8, -6) \rightarrow (8, 6)$ $\quad f(8) = \boxed{6}$

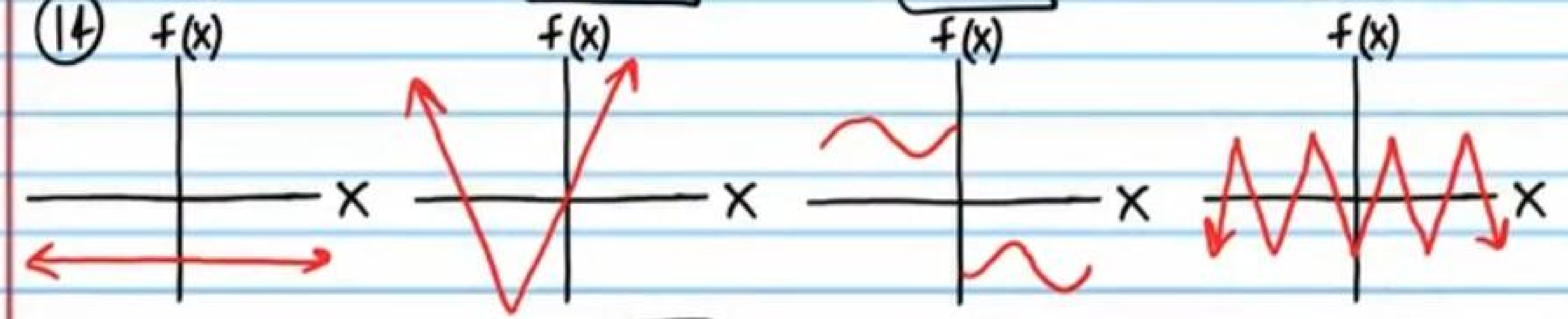
Determine if the graphs below are even functions, odd functions, or neither.

⑬



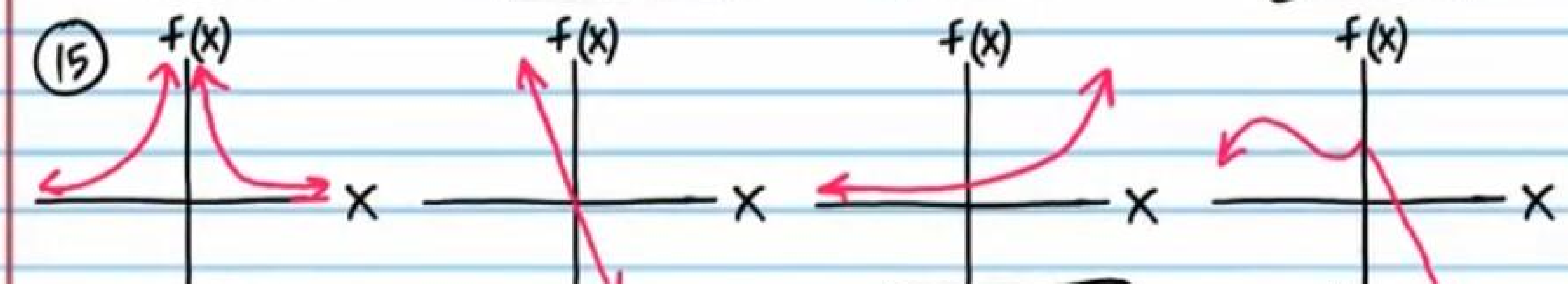
Neither | Even | Odd | Odd

⑭



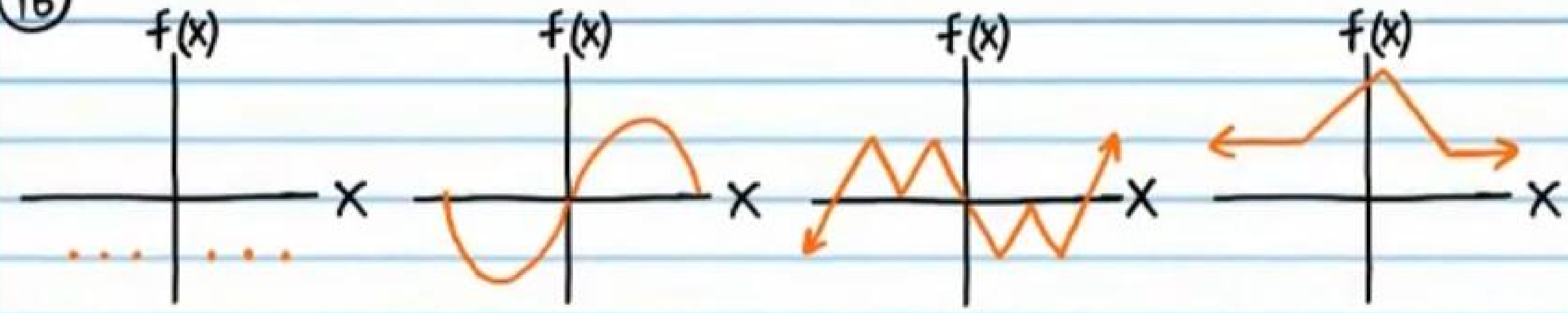
Even | Neither | Odd | Even

⑮



Even | Odd | Neither | Neither

(16)



Even | Odd | Odd | Neither

Determining if a Function Is Even Given Its Equation

Replace x with $-x$ and simplify.

Example

Relation	Function
$y = x^2+1$	$f(x) = x^2+1$
$y = (-x)^2+1$	$f(-x) = (-x)^2+1$
$y = x^2+1$	$f(-x) = x^2+1$
	$f(-x) = f(x)$
Symmetric about the y-axis	Symmetric about the y-axis (even)

Same equation

Same equation

Therefore, to determine if a function is even, we must show that $f(-x) = f(x)$.

Show that each function below is an even function.

(17) $f(x) = \ln(x^4 + |x|)$

$$f(-x) = \ln[(-x)^4 + |-x|]$$

$$f(-x) = \ln(x^4 + |x|)$$

$$f(-x) = f(x)$$

(18) $f(x) = \dfrac{1}{|x|}$

$$f(-x) = \frac{1}{|-x|}$$

$$f(-x) = \frac{1}{|x|}$$

$$f(-x) = f(x)$$

Opposite Angle Identities

$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$	$\tan(-\theta) = -\tan\theta$
$\csc(-\theta) = -\csc\theta$	$\sec(-\theta) = \sec\theta$	$\cot(-\theta) = -\cot\theta$

(19) $f(x) = \tan x^2 + \sec x$

$$f(-x) = \tan(-x)^2 + \sec(-x)$$

$$f(-x) = \tan x^2 + \sec x$$

$$f(-x) = f(x)$$

(20) $f(x) = \dfrac{\tan x}{x}$

$$f(-x) = \frac{\tan(-x)}{-x}$$

$$f(-x) = \frac{-\tan x}{-x}$$

$$f(-x) = \frac{\tan x}{x}$$

$$f(-x) = f(x)$$

(21) $f(x) = x^4 + 10x^2 + 5$

$f(-x) = (-x)^4 + 10(-x)^2 + 5$

$f(-x) = x^4 + 10x^2 + 5$

$f(-x) = f(x)$ ✓

(22) $f(x) = -\dfrac{9}{x^2}$

$f(-x) = -\dfrac{9}{(-x)^2}$

$f(-x) = -\dfrac{9}{x^2}$

$f(-x) = f(x)$ ✓

$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$	$\tan(-\theta) = -\tan\theta$
$\csc(-\theta) = -\csc\theta$	$\sec(-\theta) = \sec\theta$	$\cot(-\theta) = -\cot\theta$

(23) $f(x) = 3^{\cos x}$

$f(-x) = 3^{\cos(-x)}$

$f(-x) = 3^{\cos x}$

$f(-x) = f(x)$ ✓

(24) $f(x) = 5x^3 \csc x$

$f(-x) = 5(-x)^3 \csc(-x)$

$f(-x) = 5(-x^3)(-\csc x)$

$f(-x) = 5x^3 \csc x$

$f(-x) = f(x)$ ✓

Determining if a Function Is Odd Given Its Equation

Relation

Replace x with -x and y with -y. Then simplify.

$y = \frac{3}{2}x + x^5$

$-y = \frac{3}{2}(-x) + (-x)^5$

$-y = -\frac{3}{2}x - x^5$

$-1(-y) = (-\frac{3}{2}x - x^5)(-1)$

$y = \frac{3}{2}x + x^5$ ✓

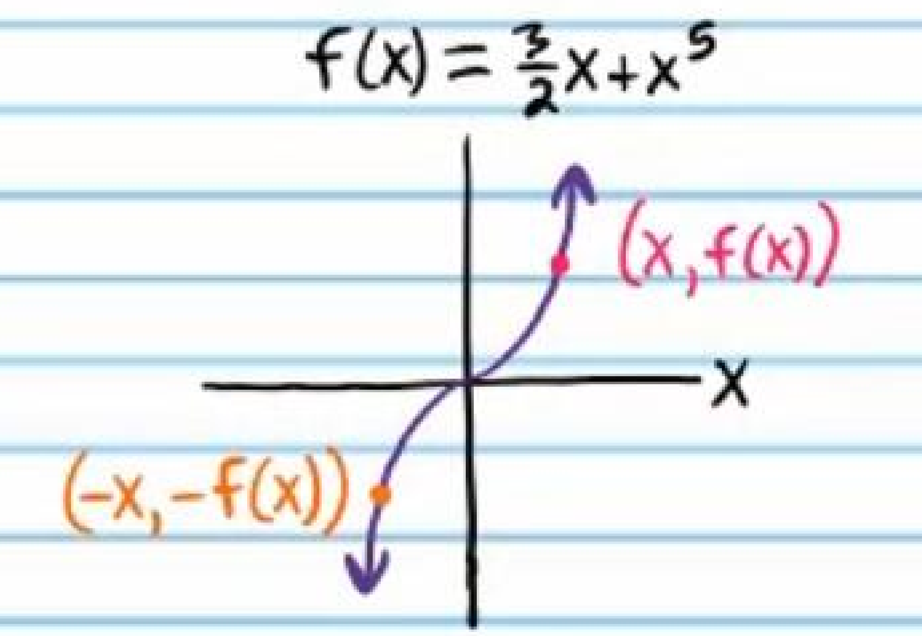
Same as the original equation

Symmetric about the origin

Function

$f(x) = \frac{3}{2}x + x^5$

We know that the function "f" is symmetric about the origin because the corresponding relation (at the left) is symmetric about the origin. Therefore, if $(x, f(x))$ is on the graph of the function, then $(-x, -f(x))$ must also be on the graph of the function.



But function notation allows us to write the point $(-x, -f(x))$ in a different way. If x is changed to $-x$, then the corresponding output (i.e. the y-value of the point) could be written as $f(-x)$. Consequently,

$(-x, -f(x)) = (-x, f(-x))$

So, if a function is symmetric about the origin (i.e. odd), then $f(-x) = -f(x)$.

The converse statement is also true

Therefore, to determine if a function is odd, we must show that $f(-x) = -f(x)$.

Prove that $f(x) = \frac{3}{2}x + x^5$ is an odd function by demonstrating that $f(-x) = -f(x)$.

$$f(x) = \frac{3}{2}x + x^5$$

$$f(-x) = \frac{3}{2}(-x) + (-x)^5$$

$$f(-x) = -\frac{3}{2}x - x^5$$

$$f(-x) = -\left(\frac{3}{2}x + x^5\right)$$

$$f(-x) = -f(x) \checkmark$$

$\sin(-\theta) = -\sin\theta$	$\cos(-\theta) = \cos\theta$	$\tan(-\theta) = -\tan\theta$
$\csc(-\theta) = -\csc\theta$	$\sec(-\theta) = \sec\theta$	$\cot(-\theta) = -\cot\theta$

Show that each function below is an odd function.

(25) $f(x) = x^3$

$$f(-x) = (-x)^3$$

$$f(-x) = -x^3$$

$$f(-x) = -f(x) \checkmark$$

(26) $f(x) = \sqrt[5]{x}$

$$f(-x) = \sqrt[5]{-x}$$

$$f(-x) = -\sqrt[5]{x}$$

$$f(-x) = -f(x) \checkmark$$

(27) $f(x) = -|x|\sin x$

$$f(-x) = -|-x|\sin(-x)$$

$$f(-x) = -|x|\sin(-x)$$

$$f(-x) = -|x|(-\sin x)$$

$$f(-x) = -(-|x|\sin x)$$

$$f(-x) = -f(x) \checkmark$$

(28) $f(x) = \sin x + \tan x$

$$f(-x) = \sin(-x) + \tan(-x)$$

$$f(-x) = -\sin x - \tan x$$

$$f(-x) = -(\sin x + \tan x)$$

$$f(-x) = -f(x) \checkmark$$

$\sin(-\theta) = -\sin\theta$ $\cos(-\theta) = \cos\theta$ $\tan(-\theta) = -\tan\theta$

$\csc(-\theta) = -\csc\theta$ $\sec(-\theta) = \sec\theta$ $\cot(-\theta) = -\cot\theta$

(29) $f(x) = \frac{x^7}{2}$

$$f(-x) = \frac{(-x)^7}{2}$$

$$f(-x) = -\frac{x^7}{2}$$

$$f(-x) = -f(x)$$

(30) $f(x) = \frac{x}{\sec x}$

$$f(-x) = \frac{-x}{\sec(-x)}$$

$$f(-x) = \frac{-x}{\sec x}$$

$$f(-x) = -\frac{x}{\sec x}$$

$$f(-x) = -f(x)$$

(31) $f(x) = -\sqrt[3]{\cot x}$

$$f(-x) = -\sqrt[3]{\cot(-x)}$$

$$f(-x) = -\sqrt[3]{-\cot x}$$

$$f(-x) = -(-\sqrt[3]{\cot x})$$

$$f(-x) = -f(x)$$

(32) $f(x) = \frac{1}{x} + x$

$$f(-x) = \frac{1}{-x} - x$$

$$f(-x) = -\left(\frac{1}{x} + x\right)$$

$$f(-x) = -f(x)$$

Show that each function below is neither odd nor even.

(33) $f(x) = 1 - x$

If even, then	If odd, then
$f(-x) = f(x)$	$f(-x) = -f(x)$
$f(-x) = 1 - x$	$f(-x) = -(1-x)$
	$f(-x) = -1 + x$

Now let's check: $f(-x) = 1 - (-x)$

$f(-x) = 1 + x$ $\boxed{\text{Neither}}$

(34) $f(x) = 2^x$

If even, then	If odd, then
$f(-x) = f(x)$	$f(-x) = -f(x)$
$f(-x) = 2^x$	$f(-x) = -2^x$

Now let's check: $f(-x) = 2^{-x}$

$\boxed{\text{Neither}}$

35. $f(x) = 9x^3 + 4x^4 + \sin x$

If even, then	If odd, then
$f(-x) = f(x)$	$f(-x) = -f(x)$
$f(-x) = 9x^3 + 4x^4 + \sin x$	$f(-x) = -(9x^3 + 4x^4 + \sin x)$
	$f(-x) = -9x^3 - 4x^4 - \sin x$

Now let's check: $f(-x) = 9(-x)^3 + 4(-x)^4 + \sin(-x)$

$f(-x) = -9x^3 + 4x^4 - \sin x$

$\boxed{\text{Neither}}$ ✓

36. $f(x) = \tan\sqrt{-x}$

If even, then	If odd, then
$f(-x) = f(x)$	$f(-x) = -f(x)$
$f(-x) = \tan\sqrt{-x}$	$f(-x) = -\tan\sqrt{-x}$

Now let's check: $f(-x) = \tan\sqrt{-(-x)}$

$f(-x) = \tan\sqrt{x}$

$\boxed{\text{Neither}}$ ✓

37. $f(x) = 2x - 1$

If even, then	If odd, then
$f(-x) = f(x)$	$f(-x) = -f(x)$
$f(-x) = 2x - 1$	$f(-x) = -(2x - 1)$
	$f(-x) = -2x + 1$

Now let's check: $f(-x) = 2(-x) - 1$

$f(-x) = -2x - 1$

$\boxed{\text{Neither}}$ ✓

38. $f(x) = \ln x$

If even, then	If odd, then
$f(-x) = f(x)$	$f(-x) = -f(x)$
$f(-x) = \ln x$	$f(-x) = -\ln x$

Now let's check: $f(-x) = \ln(-x)$

$\boxed{\text{Neither}}$ ✓

$\sin(-\theta) = -\sin\theta$ | $\cos(-\theta) = \cos\theta$ | $\tan(-\theta) = -\tan\theta$

$\csc(-\theta) = -\csc\theta$ | $\sec(-\theta) = \sec\theta$ | $\cot(-\theta) = -\cot\theta$

(39) $f(x) = x^2 + \cot x - 10$

If even, then

$f(-x) = f(x)$

$f(-x) = x^2 + \tan x - 10$

If odd, then

$f(-x) = -f(x)$

$f(-x) = -(x^2 + \tan x - 10)$

$f(-x) = -x^2 - \tan x + 10$

Now let's check: $f(-x) = (-x)^2 + \cot(-x) - 10$

$f(-x) = x^2 - \cot x - 10$

$\boxed{\text{Neither}}$ ✓

(40) $f(x) = \dfrac{1}{x^3 + x^2 + x + 1}$

If even, then

$f(-x) = f(x)$

$f(-x) = \dfrac{1}{x^3 + x^2 + x + 1}$

If odd, then

$f(-x) = -f(x)$

$f(-x) = -\dfrac{1}{x^3 + x^2 + x + 1}$

Now let's check: $f(-x) = \dfrac{1}{(-x)^3 + (-x)^2 - x + 1}$

$f(-x) = \dfrac{1}{-x^3 + x^2 - x + 1}$

$\boxed{\text{Neither}}$ ✓

Review on the Exponent Rules

Pre-Algebra Rules

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\dfrac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\dfrac{x^9}{x^3} = x^{9-3} = x^6$

Intermediate Algebra Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\dfrac{x}{y}\right)^a = \dfrac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\dfrac{n}{3}\right)^7 = \dfrac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

More Intermediate Algebra Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \dfrac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \dfrac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Review on Exponential Equations

Solve.

(41) $25^x = \frac{1}{5}$

VI $25^x = \frac{1}{5^1}$

VIII $25^x = 5^{-1}$

$(5^2)^x = 5^{-1}$

V $5^{2x} = 5^{-1}$

$2x = -1$

$\boxed{x = -\frac{1}{2}}$

(42) $\frac{3^6}{\sqrt[4]{3}} = \frac{27^x}{9^x}$

IX $\frac{3^6}{3^{\frac{1}{4}}} = \frac{27^x}{9^x}$

IV $\frac{3^6}{3^{\frac{1}{4}}} = \left(\frac{27}{9}\right)^x$

II $3^{6-\frac{1}{4}} = 3^x$

$3^{\frac{23}{4}} = 3^x$

$\boxed{\frac{23}{4} = x}$

(43) $\sqrt[5]{2^x}\cdot 10^{\frac{x}{5}} = 1$

X $2^{\frac{x}{5}}\cdot 10^{\frac{x}{5}} = 1$

III $(2\cdot 10)^{\frac{x}{5}} = 1$

$20^{\frac{x}{5}} = 1$

VIII $20^{\frac{x}{5}} = 20^0$

$\frac{x}{5} = 0$

$\boxed{x = 0}$

(44) $\frac{1}{4} = 16^x 4^x$

VI $\frac{1}{4^1} = 16^x 4^x$

VIII $4^{-1} = 16^x 4^x$

III $4^{-1} = (16\cdot 4)^x$

$4^{-1} = 64^x$

$4^{-1} = (4^3)^x$

V $4^{-1} = 4^{3x}$

$-1 = 3x$

$\boxed{x = -\frac{1}{3}}$

(45) $\frac{25^{x+2}}{5^{x+2}} = \frac{\sqrt[6]{5}}{125}$

IV $\left(\frac{25}{5}\right)^{x+2} = \frac{\sqrt[6]{5}}{125}$

IX $5^{x+2} = \frac{5^{\frac{1}{6}}}{5^3}$

II $5^{x+2} = 5^{\frac{1}{6}-3}$

$5^{x+2} = 5^{-\frac{17}{6}}$

$x+2 = -\frac{17}{6}$

$\boxed{x = -\frac{29}{6}}$

(46) $1 = (\sqrt[3]{7})^x\cdot 7^{5x-5}$

VIII $7^0 = (\sqrt[3]{7})^x\cdot 7^{5x-5}$

X $7^0 = 7^{\frac{x}{3}}\cdot 7^{5x-5}$

I $7^0 = 7^{\frac{x}{3}+5x-5}$

$7^0 = 7^{\frac{16x}{3}-5}$

$0 = \frac{16x}{3} - 5$

$\boxed{x = \frac{15}{16}}$

Review on Logarithms

Example

Solve: $3^x = 9$ Do not need a calculator.

Example

Solve: $3^x = 10$ Need a calculator.

Notation for Calculating the Value of Exponents

If $b^x=c$, then enter $\log_b c$ into your calculator to find x.

Note: Traditionally, neither "c" nor "b" is allowed to be negative or zero when placed in a logarithmic expression.

Write the solutions of the equations below using "$\log_b c$" notation. Use a calculator to approximate (i.e. Round) your answers to three decimal places.

(47) $5^x=9$

$x=\log_5 9$

$\boxed{x \approx 1.365}$

(48) $2=3^x$

$x=\log_3 2$

$\boxed{x \approx 0.631}$

(49) $17^x=8$

$x=\log_{17} 8$

$\boxed{x \approx 0.734}$

(50) $100=91^x$

$x=\log_{91} 100$

$\boxed{x \approx 1.021}$

(51) $7^x=6$

$x=\log_7 6$

$\boxed{x \approx 0.921}$

(52) $20=6^x$

$x=\log_6 20$

$\boxed{x \approx 1.672}$

(53) $8^x=10$

$x=\log_8 10$

$\boxed{x \approx 1.107}$

(54) $5=2^x$

$x=\log_2 5$

$\boxed{x \approx 2.322}$

More Exponential Equations

Determine if you can solve the following equations without a calculator. If you need a calculator, write the solution in $\log_b c$ notation, and then use a calculator to approximate the answer to the nearest ten thousandth.

(55) $2(6^x)+9=81$

$2(6^x)=72$

$\frac{2(6^x)}{2}=\frac{72}{2}$

$6^x=36$

$6^x=6^2$

$x=2$

No Calculator

(56) $-1=\frac{10^x}{5}-3$

$2=\frac{10^x}{5}$

$(5)2=\frac{10^x}{5}(5)$

$10=10^x$

$10^1=10^x$

$x=1$

No Calculator

(57) $-8=-4(8^x)+4$

$-12=-4(8^x)$

$\frac{-12}{-4}=\frac{-4(8^x)}{-4}$

$3=8^x$

$x=\log_8 3$

$x \approx 0.5283$

Calculator

VI

(58) $1+\frac{5^x}{2}=1.5$

$\frac{5^x}{2}=0.5$

$5^x=1$

VII $5^x=5^0$

$\boxed{x=0}$ No calculator

(59) $2(9^x)+6=22$

$2(9^x)=16$

$9^x=8$

$x=\log_9 8$

$\boxed{x\approx 0.9464}$ Calculator

(60) $32=7(3^x)-10$

$42=7(3^x)$

$6=3^x$

$x=\log_3 6$

$\boxed{x\approx 1.6309}$ Calculator

Review on Logarithmic Equations

Solve.

(61) $3\log_2 x-8=1$

$3\log_2 x=9$

$\frac{3\log_2 x}{3}=\frac{9}{3}$

$\log_2 x=3$

$2^3=x$

$\boxed{8=x}$

(62) $6=\frac{\log_x 5}{2}+1$

$5=\frac{\log_x 5}{2}$

$(2)5=\frac{\log_x 5}{2}(2)$

$10=\log_x 5$

$x^{10}=5$

$\sqrt[10]{x^{10}}=\pm\sqrt[10]{5}$

$x=\pm\sqrt[10]{5}$

$\boxed{x=\sqrt[10]{5}}$

(63) $4 = 7\log_x 9 + 3$

$1 = 7\log_x 9$

$\frac{1}{7} = \log_x 9$

$x^{\frac{1}{7}} = 9$

IX $\sqrt[7]{x} = 9$

$(\sqrt[7]{x})^7 = (9)^7$

$\boxed{x = 9^7}$

(64) $-6 - \frac{\log_4 x}{3} = -5$

$-\frac{\log_4 x}{3} = 1$

$\log_4 x = -3$

$4^{-3} = x$

VIII $\frac{1}{4^3} = x$

$\boxed{\frac{1}{64} = x}$

(65) $10 + 4\log_6 x = 14$

$4\log_6 x = 4$

$\log_6 x = 1$

$6^1 = x$

VI $\boxed{6 = x}$

(66) $\frac{\log_x 81}{4} - 11 = -10$

$\frac{\log_x 81}{4} = 1$

$\log_x 81 = 4$

$x^4 = 81$

$\sqrt[4]{x^4} = \pm\sqrt[4]{81}$

$x = \pm 3$

$\boxed{x = 3}$

(67) $17 = 32\log_x 8 + 1$

$16 = 32\log_x 8$

$\frac{1}{2} = \log_x 8$

$x^{\frac{1}{2}} = 8$

IX $\sqrt{x} = 8$

$(\sqrt{x})^2 = (8)^2$

$\boxed{x = 64}$

(68) $\frac{\log_7 x}{6} + 5 = 5$

$\frac{\log_7 x}{6} = 0$

$\log_7 x = 0$

$7^0 = x$

VII $\boxed{1 = x}$